

ALICE

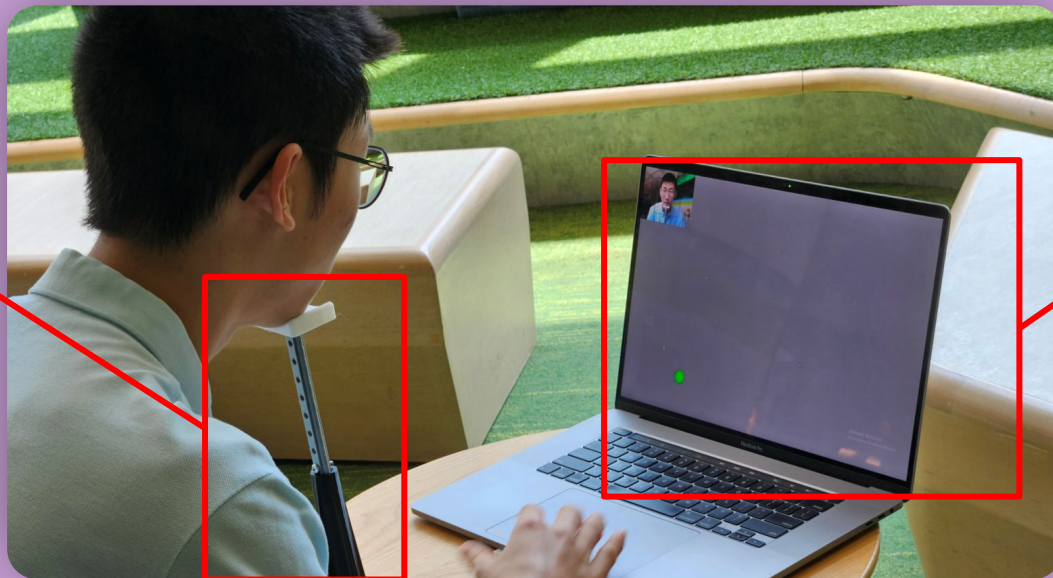
Alzheimer Diagnosis and Severity
Indicator based on Eye Behaviour Neural
Network Classifier



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ALICE

ALEX MK1

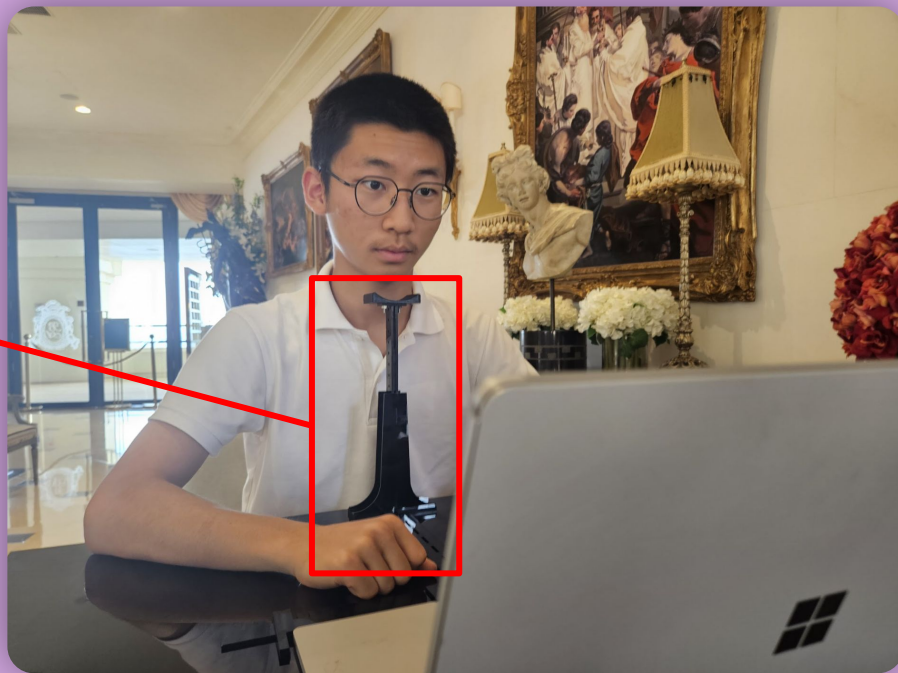


ALICE
(Data collection
utility shown)

ALICE: Alzheimer's diagnosis and severity IndiCator based on Eye behavior neural network classifier
ALEX: Alzheimer's diagnosis EXamination support

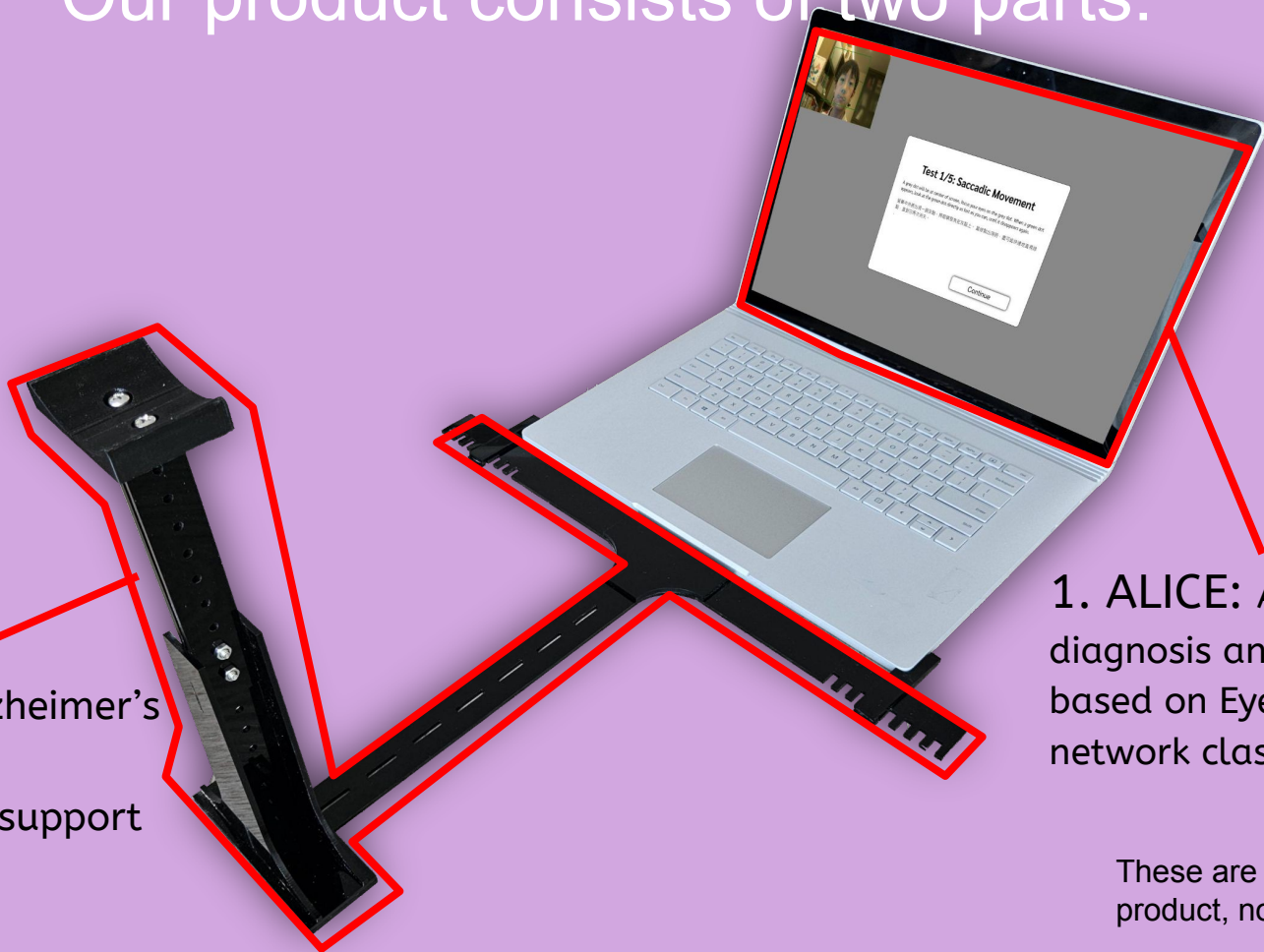
ALICE

ALEX MK2



ALICE: Alzheimer's diagnosis and severity IndiCator based on Eye behavior neural network classifier
ALEX: Alzheimer's diagnosis EXamination support

Our product consists of two parts:



1. ALICE: Alzheimer's diagnosis and severity Indicator based on Eye behavior neural network classifier

These are photos of the actual product, not a 3D render

2. ALEX: Alzheimer's EXamination support

Technical Proof of concept (PoC) - ALICE

```

<div>
  <body>
    <script class="test1">
      //this is the center dot
      greenDot.className = 'green-dot';

      canvas.appendChild(centerDot);
      canvas.appendChild(greenDot);

      webgazer.setGazeListener(function(data, elapsedTime) {
        if (data == null) {
          return;
        }
        var xprediction = data.x; //these x coordinates are relative to the viewport
        var yprediction = data.y; //these y coordinates are relative to the viewport
      }).begin();
      function lol() {
      }
      function updateGreenDotPosition() {
        if (counter == 0) {
          textbox = document.getElementsByClassName("tempdisplay")[0];
          textbox.style.display="block";
          textbox.getElementsByClassName("error")[0].textContent = "This test has finished. Click Continue to go to the next test";
          textbox.getElementsByClassName("yes")[0].textContent = "Continue";
          textbox.getElementsByClassName("yes")[0].onclick = why;
        }
        const currentTime = Date.now();
        const randomAngle = Math.random() * 2 * Math.PI;
        const newX = ( Math.random() - 0.50 ) * 50 ;
        const newY = ( Math.random() - 0.50 ) * 50 ;

        greenDot.style.left = `calc(50% + ${newX}%)`;
      }
    </script>
  </body>
</div>

```

Code for:

- Tracking and logging eye movement
- sample code for code for test 1

```
button:hover {
    background-color: #ddd;
    scale: 1.05;
}

button:active {
    background-color: #eee;
    scale: 0.95;
}

.onTop {
    position: absolute;
```

Code for:

- Styling of website for simple but understandable GUI

```

325 function draw() {
326   ctx.moveTo(width/2, height/2 - fixationCrossHeight);
327   ctx.lineTo(width/2, height/2 + fixationCrossHeight);
328   ctx.fillStyle = fixationCrossColor;
329   ctx.stroke();
330 }
331
332 //Update the dots with their new location
333 function updateDots() {
334   //Cycle through to the next set of dots
335   if (currentSet == nSets - 1) {
336     currentSet = 0;
337   } else {
338     currentSet++;
339   }
340
341   //Load in the current set of dot array for easy handling
342   dotArray = dotArray2[currentSet]; //Global variable, so the draw function also uses this array
343
344   //Loop through the dots one by one and update them accordingly
345   for (var i = 0; i < nDots; i++) {

```

Code for:

- Test 5: Logging and creating interface for random dot kinematics

Total 45,311 lines of code

Technical Proof of concept (PoC) - ALEX



ALEX prototype headstand testing

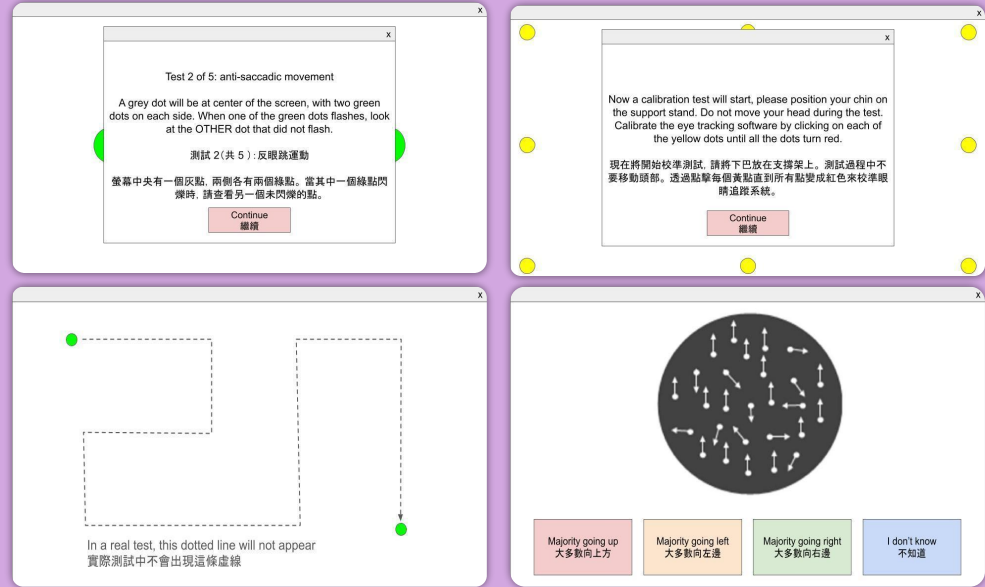
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Minimum Viable Product (MVP) - ALICE



Data collection utility

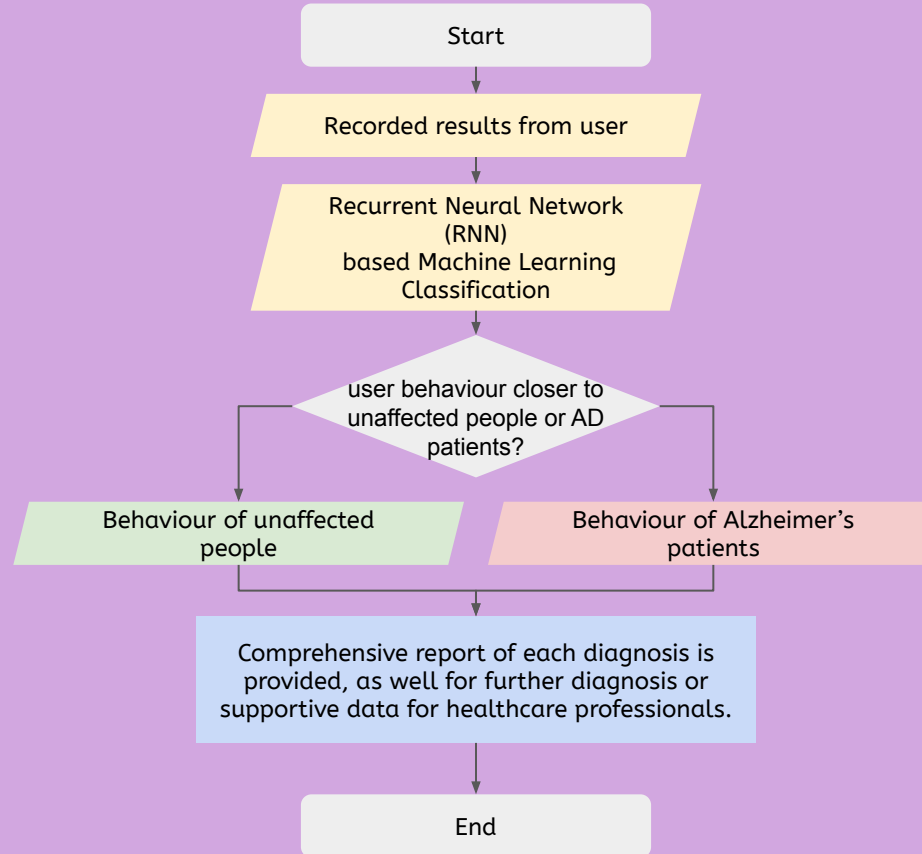
Fully functioning data collection utility software



User Interface

Fully completed user interface

ALICE - Innovative & Patentable Technology

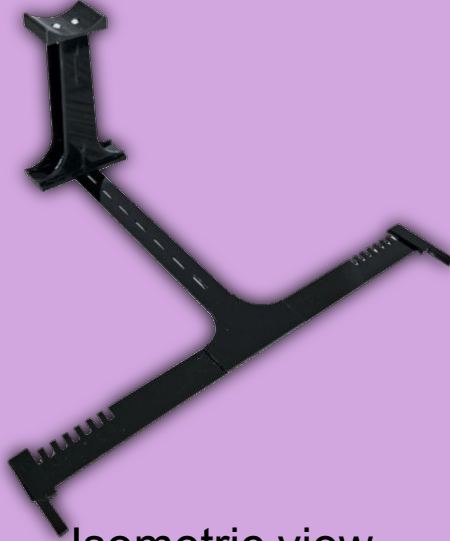


Minimum Viable Product (MVP) - ALEX

These are photos of the actual MVP, not a 3D render



Back view



Isometric view



Top view

Manufactured full sized sample of third iteration of production model for ALEX
(ALEX: Alzheimer's diagnosis EXamination support)

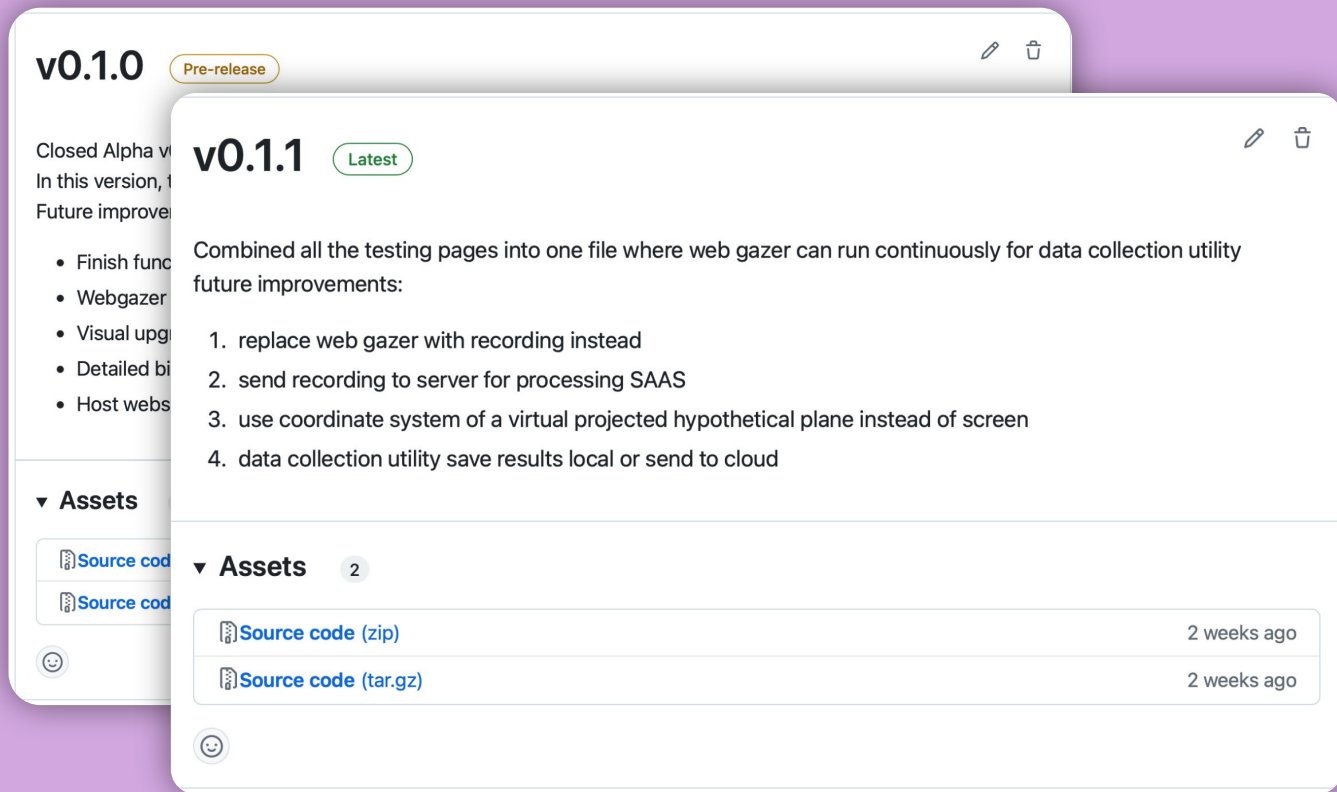
Current State of Products

1. ALICE is in MVP stage as now it fulfills all functions of the data collection utility tool and partially fulfills all the final requirements.
2. ALEX is in MVP stage as now it fulfills all functions as the design intended.

Design for Manufacturing / Pre-production Engineering / UAT

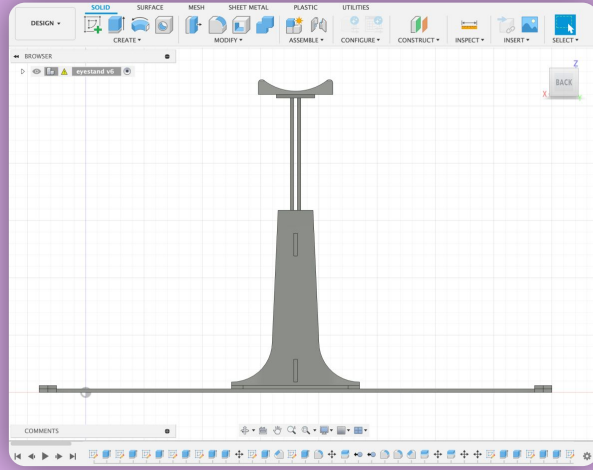
1. ALICE is software, thus requiring no infrastructure for manufacturing. It is ready for distribution at any time. The source code is maintained on github.
2. ALEX is hardware, thus requiring industrial equipment for manufacturing. It is ready for mass production. The 3D and 2D CAD designs are available.

Design for Manufacturing / Pre-production Engineering/ UAT

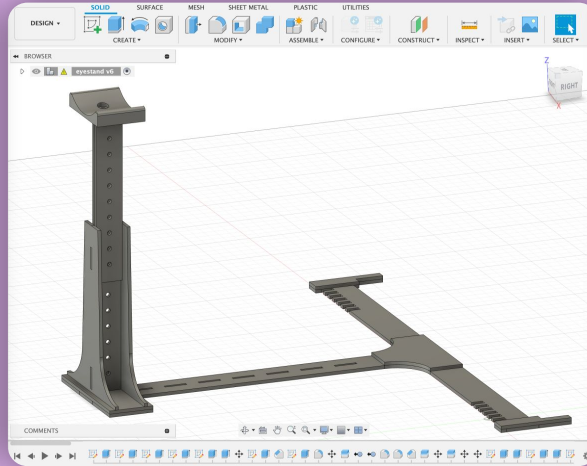


Latest version of ALICE obtainable through Github Releases

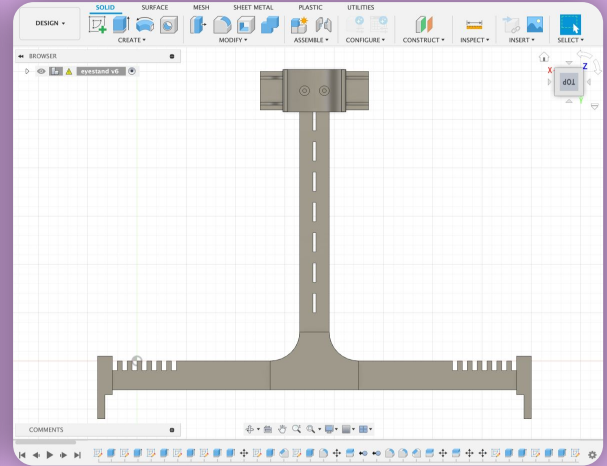
Design for Manufacturing / Pre-production Engineering / UAT



Back view



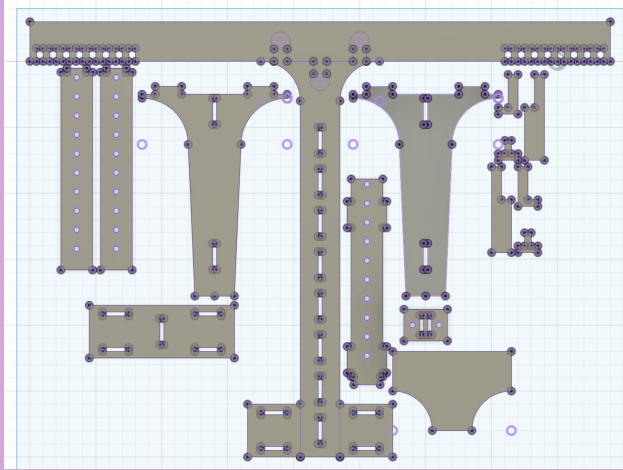
Isometric view



Top view

Manufactured full sized sample of third iteration of production model for ALEX (ALEX: Alzheimer's diagnosis EXamination support)

Design for Manufacturing / Pre-production Engineering / UAT



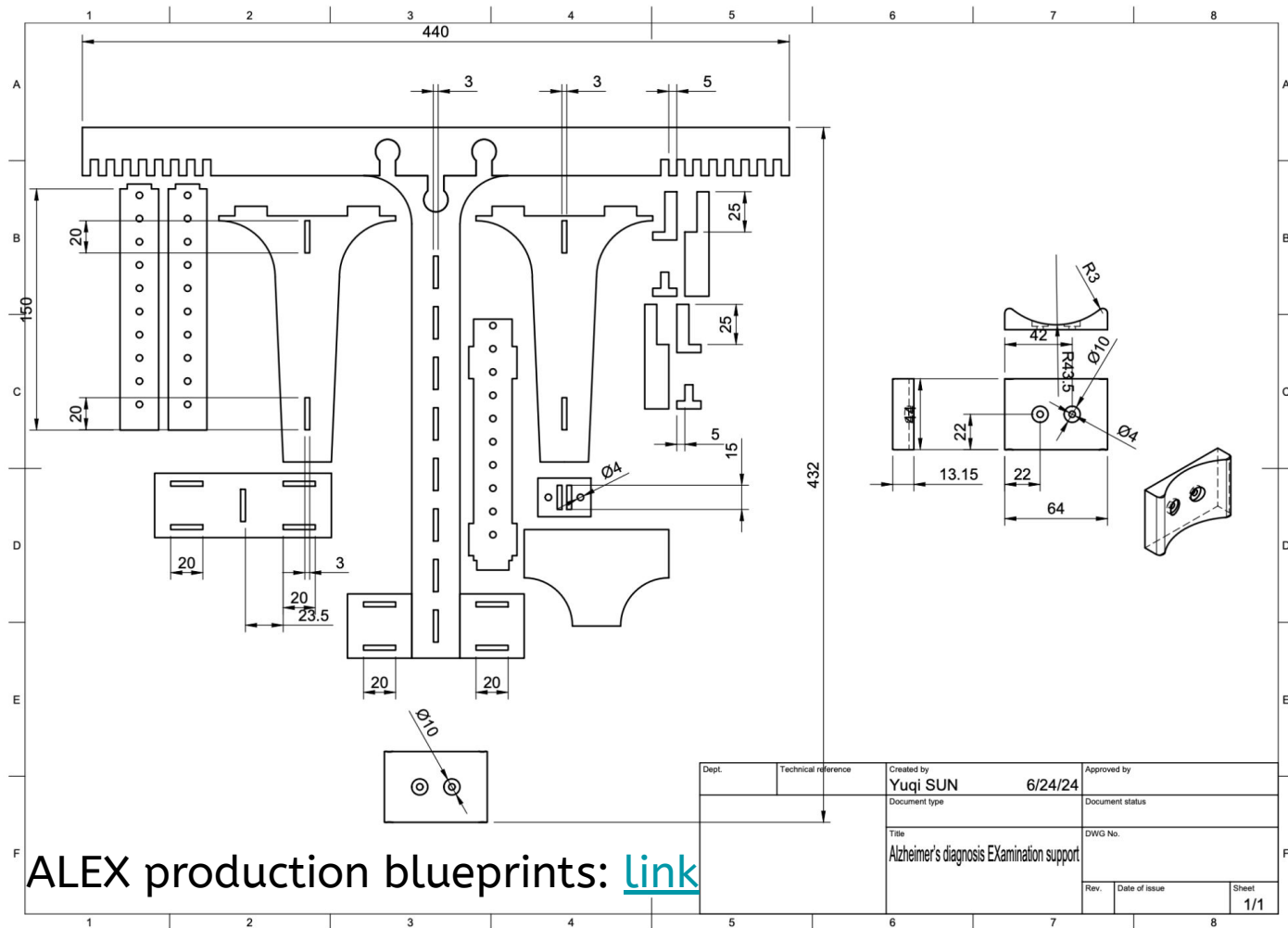
1:1 Blueprint

Video demo: <https://youtu.be/-vH4L5KMExE>



Eye stand testing

The CAD designs for ALEX (Alzheimer's diagnosis EXamination support) are ready for the third iteration of production



ALEX production blueprints: [link](#)